

Abim Ferdous

(562)-574-9446 | abimferdous@gmail.com | <https://abim-portfolio.onrender.com/>

EDUCATION

Cerritos College, 11110 Alondra Blvd, Norwalk, CA
Intended Major: Mechanical Engineering (2nd year)

Expected Graduation Year 2027
GPA: 4.0

PROJECTS

EduGotchi (UCLA IEEE Hackathon, top 10)

- Developed EduGotchi, an AI-driven educational companion designed to increase early childhood learning.
- Features integrated Agentic AI via ChatGPT API to dynamically generate and adapt based on user.
- Engineered an interactive workflow that blends AI with gamified educational content.

SolidWorks Projects

- Currently working on a 3DOF robotic arm using nema 17 stepper motors for manual control via a web interface, with autonomous operation planned as a future phase.
- Calculated and evaluated a 4:1 planetary gear configuration for joint reduction, identifying structural limitations for shoulder's load requirements.
- Investigating cycloidal drive alternatives to address the shoulder joint's torque and durability needs.

SKILLS

Programming

- Python, C++, HTML and CSS.

CAD

- OnShape, Solidworks (CSWA certified), <https://tinyurl.com/2p23rm4k>.

Electrical Skills

- PCB designing using EasyEDA.
- Soldering and circuit connection through IEEE @ Cerritos College in partnership with UCLA IEEE.

EXPERIENCE

UCI Rocketry Summer Camp

314 Los Trancos Dr, Irvine, CA 92617

NAR LEVEL 1 CERTIFICATION

August 3rd-August 8th 2026

- Earned the NAR LEVEL 1 certification after designing, building and launching a rocket to a height of 2500 ft.
- Simulated using NASA's Rocket Modeler III to iterate nose cone shape, body length, fin geometry and Cg/Cp positions.
- Learned and applied theory from thermodynamics and aerodynamics to predict rocket behaviour.

USC INKPEN LAB INTERN

700 Childs Way, Los Angeles, CA 90089-0911

Working on Single Molecular Electronics (SME)

June 2026-July 2026

- Engineered analysis code for the INKPEN Lab to process Osmium(III) molecular junction data, creating a standardized long-term asset for molecular electronics research.
- Designed using IGOR PRO a custom data-filtering algorithm to eliminate experimental noise to isolate and visualize current-voltage (I-V) characteristics of Osmium (III) complexes.
- Maintained physical laboratory apparatus used for the raw data collection while ensuring hardware is reliable and data is read correctly.

Vex Robotics Lead Programmer

11110 Alondra Blvd, Norwalk, CA, 90650

Competition at multiple universities

November 2025-February 2026

- Developed a C++ path-planning algorithm for the autonomous period, navigating field constraints to maximize scoring.
- Facilitated design-to-production workflows, coordinating with 15+ members across CAD, physical build, and code.
- Secured top placement at multiple regional tournaments, qualifying the team for VEX U Worlds.

Design Engineer for NCAS with NASA

13111 Sycamore Dr, Norwalk, CA, 90650

At DoubleTree by Hilton Hotel

January 8-January 12 2026

- Modeled competition-ready rovers that successfully collected mineral samples every round while navigating terrain.
- Collaborated with a 12-person team to deliver rover designs within strict time and budget constraints.
- Iterated designs across rounds to maintain scoring performance under changing terrain conditions.